Types of Relationships in Database

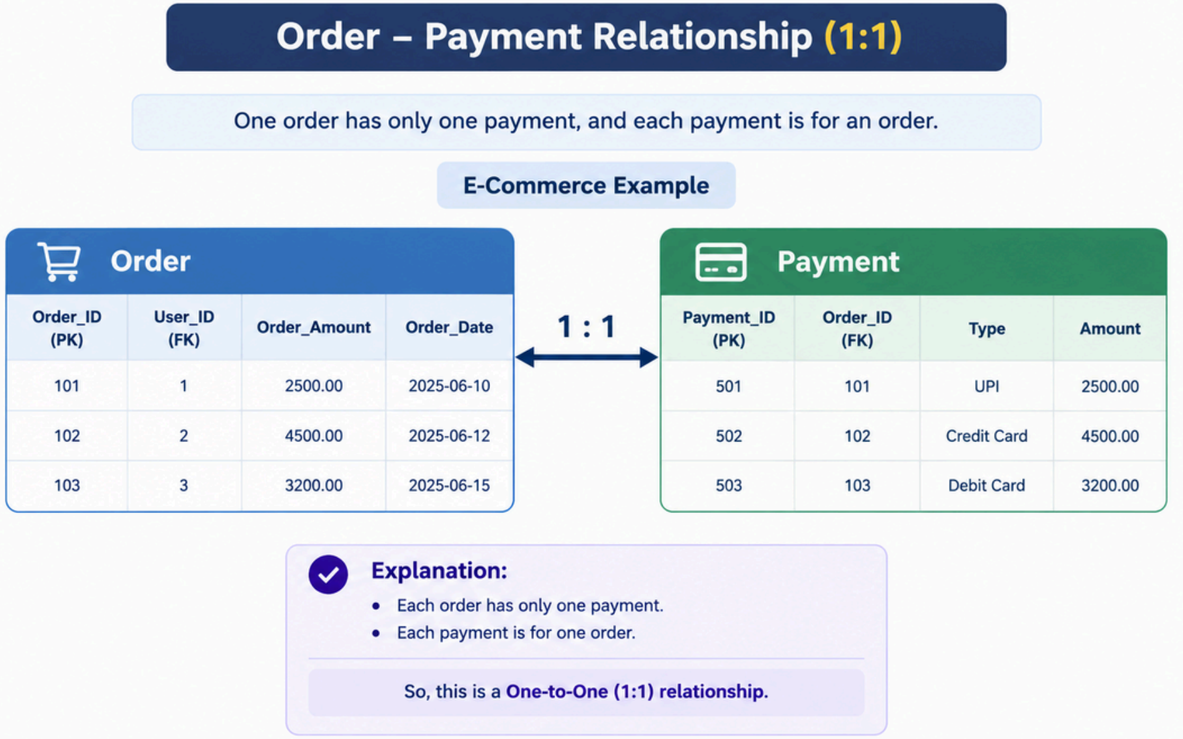
One-to-One Relationships

Many-to-One Relationships

Many-to-Many Relationships

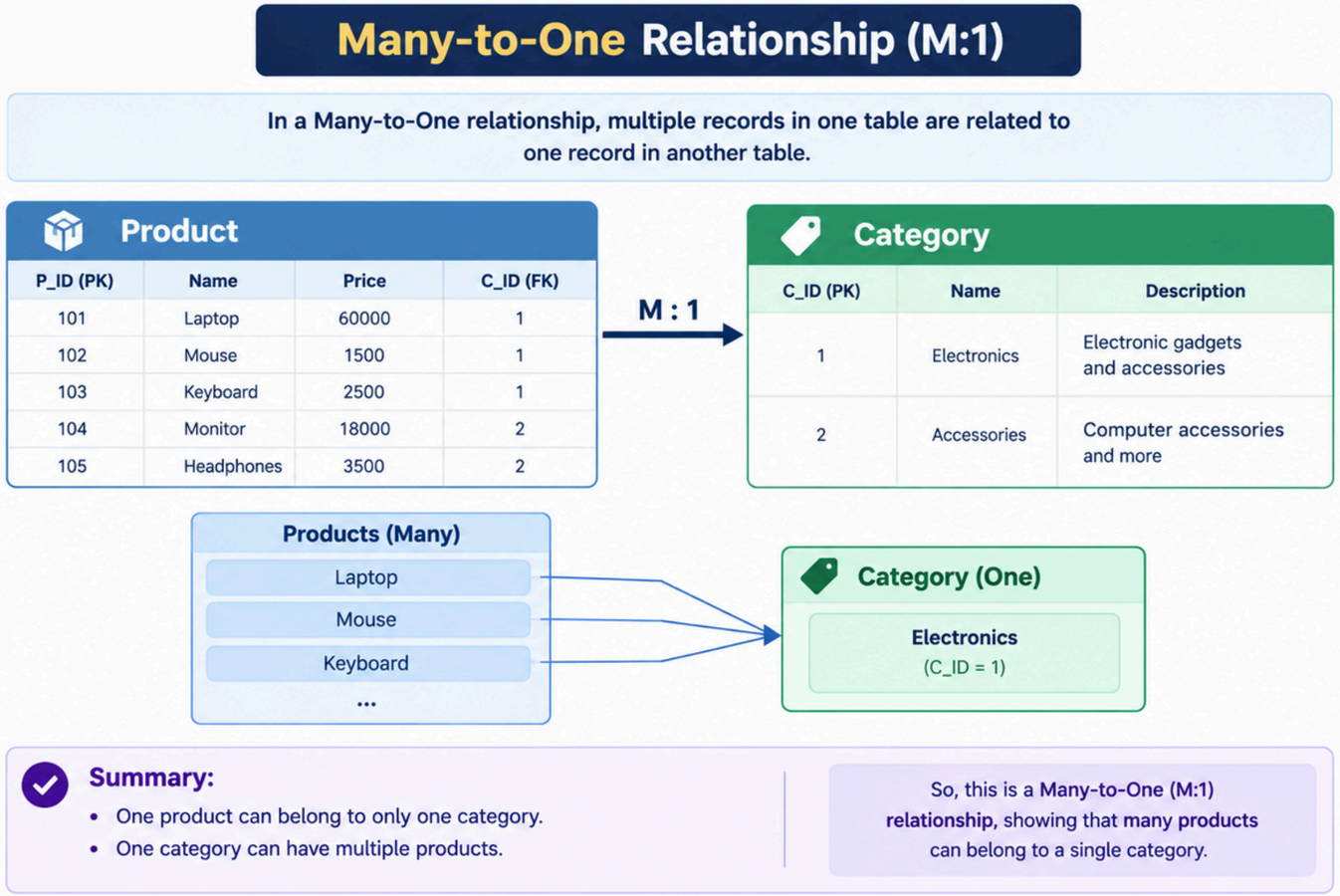
One-to-One Relationships (1:1)

In a One-to-One relationship, one record in a table is related to only one record in another table, and that record is also related to only one record in the first table.



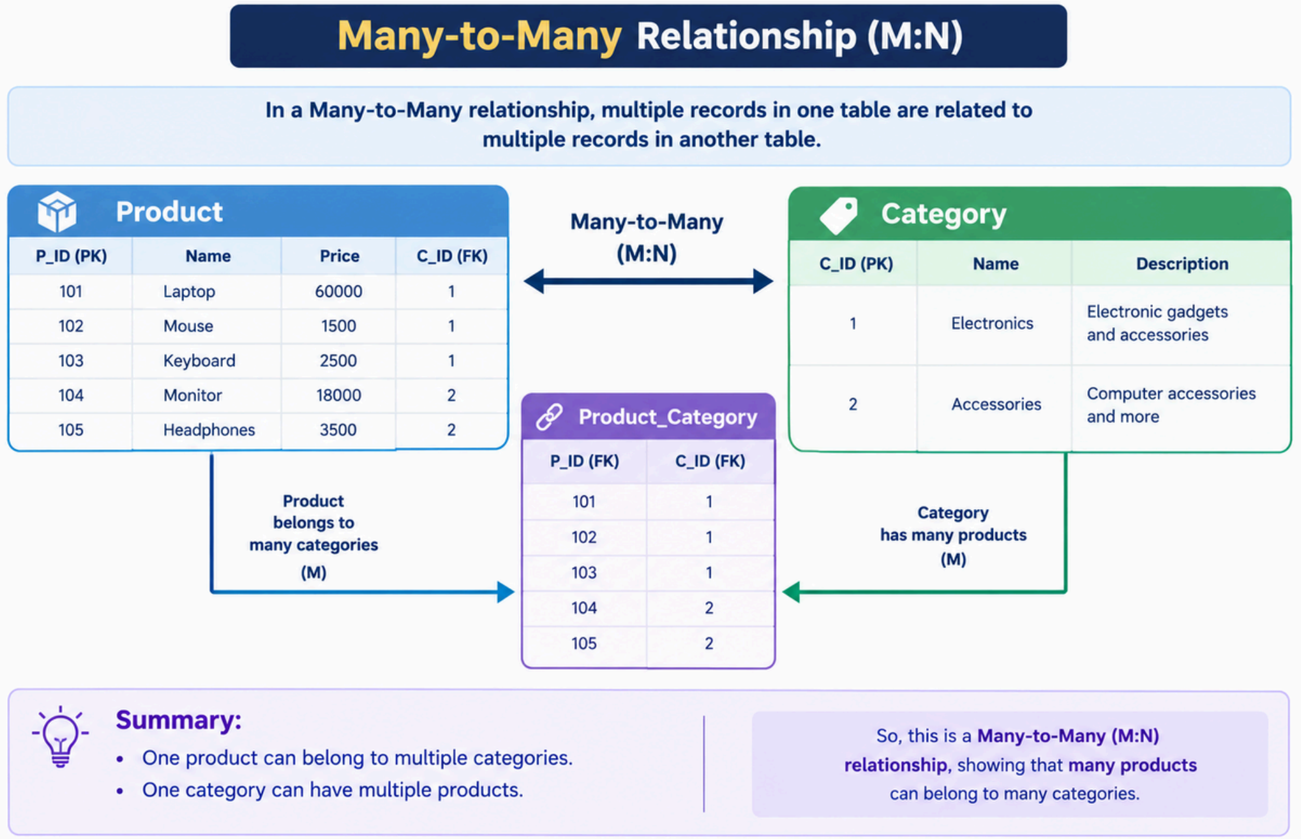
One-to-Many Relationship (1:M)

In a One-to-Many relationship, one record in a table is related to multiple records in another table, while each record in the second table is related to only one record in the first table.

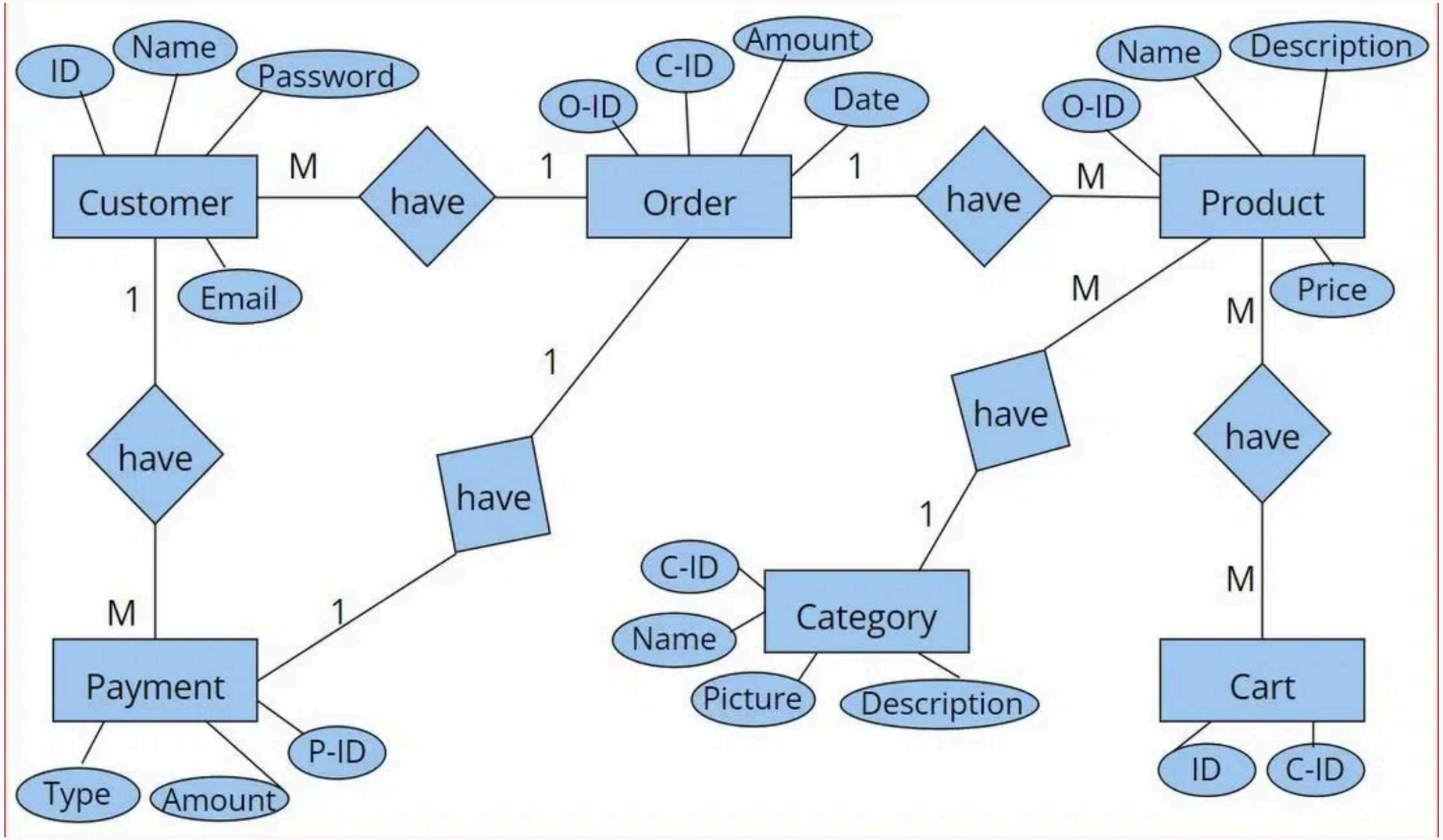


Many-to-Many Relationship (M:M)

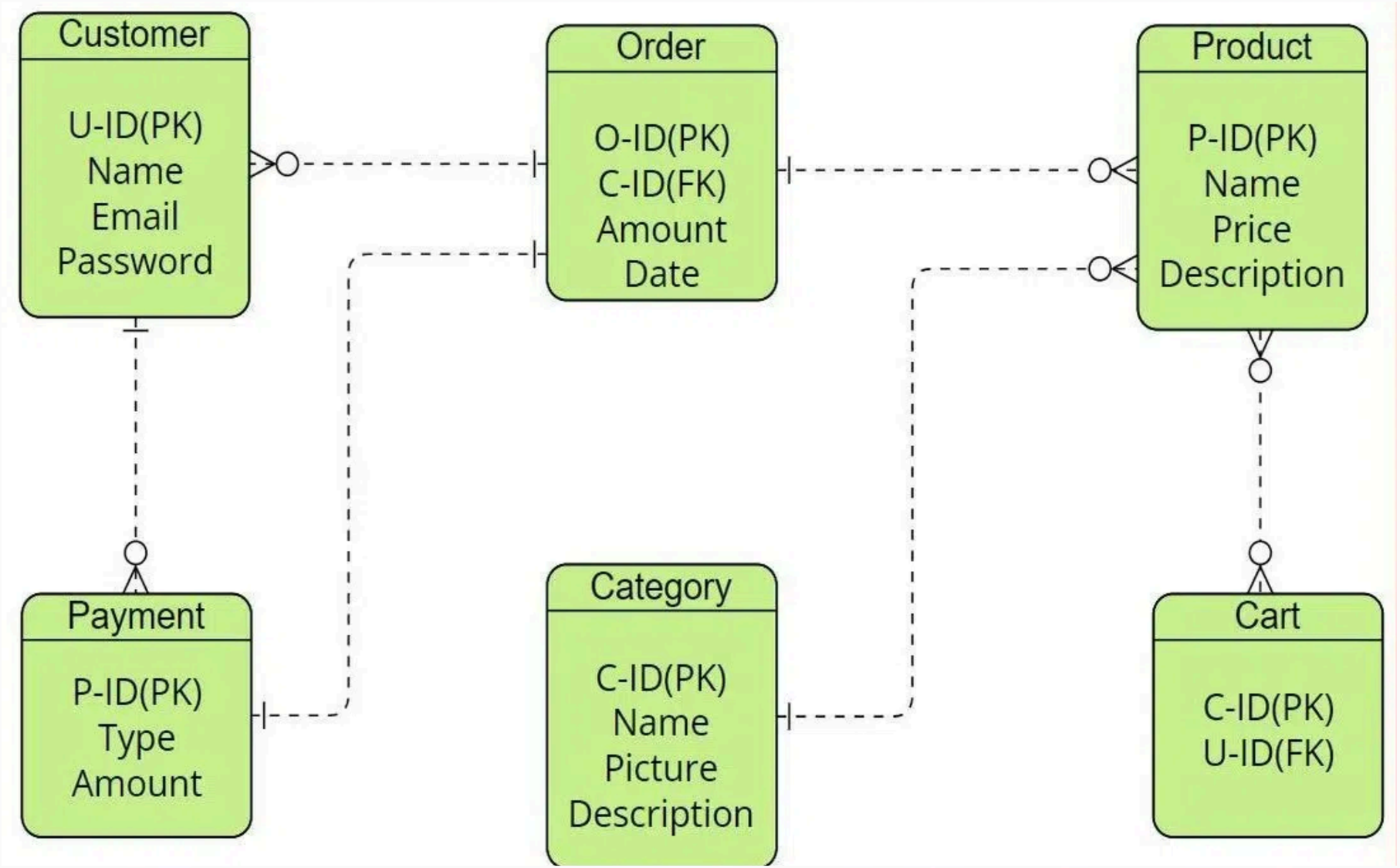
In a One-to-Many relationship, one record in a table is related to multiple records in another table, while each record in the second table is related to only one record in the first table.



Representation of ER Diagram



Representation Database Model of E-commerce



AUTO_INCREMENT start

```
CREATE TABLE Product (
   P_ID INT PRIMARY KEY AUTO_INCREMENT,
   Name VARCHAR(255) NOT NULL UNIQUE,
   Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
   Description TEXT NOT NULL
)AUTO_INCREMENT = 10011;

CREATE TABLE Orders (
   Order_ID INT PRIMARY KEY AUTO_INCREMENT,
   Order_Amount DECIMAL(10,2) NOT NULL CHECK(Order_Amount > 0),
   Order_Date DATE NOT NULL
   P_ID INT NOT NULL,
   FOREIGN KEY (P_ID) REFERENCES Product(P_ID)
)AUTO_INCREMENT = 10011;
```

ALTER TABLE with AUTO_INCREMENT

Method 1: After table create

```
ALTER TABLE Product AUTO_INCREMENT = 10011;
```

Explanation:

- ALTER TABLE Product → existing table ko modify karna.
- AUTO_INCREMENT = 1011 → next automatically generated ID 1011

Method 2: Existing IDs change

```
UPDATE Product
SET P_ID = P_ID + 1010;

UPDATE Product
SET P_ID = 1011
WHERE P_ID = 1;
```

Multiple IDS change

```
UPDATE Product
SET P_ID = P_ID + 1010
WHERE P_ID BETWEEN 1 AND 4;
```

How to Rollback your work in MySQL ?

Ecommerce Product Table

P_ID	Name	Price	Stock	Category
10011	Laptop	85000	15	Electronics
10012	Smart Watch	5200	25	Accessories
10013	Wireless Mouse	2500	40	Accessories
10014	Gaming Keyboard	6300	18	Accessories

6000

SET Price = 6000

This update changes the price of P_ID = 10012 (Smart Watch) to 6000.

```
UPDATE product
SET Price = 6000
WHERE P_ID = 10012;
```

COMMIT;

If you run **COMMIT**, the change will be permanent and saved in the database.

ROLLBACK;

If you run **ROLLBACK**, the change will be undone and the price will return to 5200.

COMMIT & ROLLBACK WORK FOR:

- Insert
- Update
- Delete


Thank
you